Heart Plaque Detection System Design using a ML Technique "Naive Bayes and LSSVM"

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***Abstract*— It is generally acknowledged that the prediction of cardiovascular disease is essential for informing practitioners and hence allowing educated health-related decisions to take place. Early identification is algorithmically seen to motivate lifestyle changes if appropriate and also ensure ideal medical therapy if needed in the population. Given that early intervention is known to be beneficial, machine learning (ML) appears as a valuable tool for predicting and gaining insight into heart disease symptoms. Our study adds to this knowledge base as it demonstrates that the Naive Bayes algorithm was a better predictor of cardiac plaque disease than the LSSVM method. The goal of the research work is to conduct heart plaque diagnosis using Naive Bayes algorithm and compare the accuracy with Least Squares SVM algorithm. The study divides the Naive Bayes method and the Least Squares SVM algorithm into two separate groups, with 20 samples being used to derive G power from the Kaggle dataset from the heart plaque disease depending on Clinical with a 95% confidence interval. LSSVM Algorithm has achieved accuracy of only 67.3%, While Naive Bayes Algorithm gives the best accuracy of 78%. The dataset of this study found that the LSSVM algorithm had the lowest performance when compared to Naive Bayes-algorithm for detecting heart plaque.**

Keywords- Heart plaque detection, Naive Bayes algorithm, Least Squares SVM algorithm, accuracy, Kaggle dataset, Clincalc, training set, test set.

I. Introduction

This study explores the approaches used to detect, diagnose, and self-manage Heart Plaque Disease. The presence of plaques in the heart and their natures and options for self-detection and control of Coronary Disease were systematically evaluated. This study explores a machine learning based prediction system to find the best classifier that provides the best result against clinical outcomes[1].

Naive Bayes-algorithm:

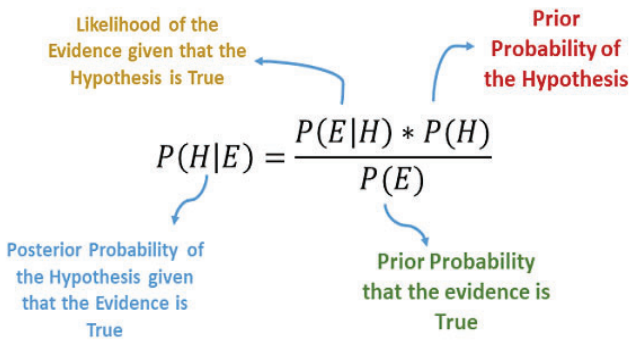

Fig. 1. Naive Bayes -algorithm

One suggested method is predictive analysis which can be used to determine the features associated with heart plaque formation which may have been considered predictive of the disease in the past, in order to detect plaque formation at an earlier stage [2]. Another approach the researchers have tried, as many rr, that report accuracies. But, scientists are always in search for some new classifiers, or combining multiple of them to increase their models accuracy. Materials and Methods: This study therefore proposes an analysis to infer based on general patient diagnostics data the algorithm that is most optimal in identifying current disease and predicting the development of heart plaque in the future.

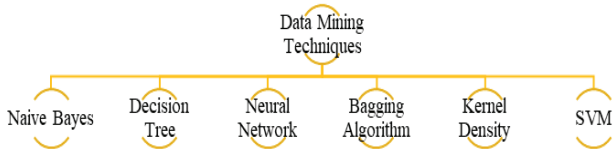

Fig. 2. Data Minings techniques to detect Heart diseases

Figure 1: Machine-learning methods for Heart Plaque disease Recent research has been conducted on different heart plaques disease diagnoses. In detecting cardiac disease, data mining is critical to that. The massive flower of mining medical records indicates huge promise in disclosing hidden patterns applicable other than clinical diagnosis of any diseases in any databases. Various data mining techniques are available to diagnose heart disease: Naive Bayes, Decision Trees, Neural Networks, Bagging Algorithms, Kernel Density and Support Vector Machines. Different procedures have different accuracy levels. Naive Bayes is a fast and efficient classification technique and is commonly used to predict whether a person has a heart problem or not.In preparation for all of this, you read the journal paper and you answer the questions (below) scrawling in Web data.

Publication on heart plaque disease published in scientific journals during last five years research articles were 7, were found publication in Google scholar 547. For example, in their recent paper: Analysis of Heart Plaque for Early Prediction Using Wavelet Transform Algorithm [3], they predicted the heart plaque with a 72 percent accuracy using the Wavelet Transform approach. The author [4] obtained the most valuable features from the Heart Plaque disease patient and reported the accuracy of 73% for identification of the disease. Detection heart plaque disease, algorithms like CWT and Huygens had been trained by [5] and [6] researchers and an 80 percent accuracy PCA, MRMR along five cross-validations were proposed by [7] author for the modeling analysis for

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deduction of the dimensionalities were proposed and an accuracy of 67 percent was obtained. Using wavelet transform algorithms, researchers also achieved a diagnosis for Heart Plaque disease with 75.7% accuracy [8].

The sorting strategy of a new supervision technique should be combined with three methods: CFS+Filter Subset Eval [9]. It turned out to be 85.5 percent accurate. A combination of k-means clustering and Naive Bayes was proposed in [10]. To improve the diagnostic performance of Naive Bayes for the heart disease patients, we did centroid initialization for Naive Bayes. They achieved an accuracy of 84.5%. Some of them The Heart Disease Prediction System (HDPS) is a decision support system based on Naive Bayesian classification and Jelinek-Mercer smoothing technique work presented in [11] Laplace smoothing gives you an approximate function designed to isolate key patterns in the data and filter out noise. It has an 86% accuracy rate on that function. In[12] a determinator involved in both the statistically based Naive Bayes determinator (cNK) and also the distance-based K-entermost neighbor. When used in a classifier, it achieved 85.92%accuracy on a heart disease dataset.

So, it returns the incorrect results as the given approach is sensitive to outliers. The main objective is to improve the accuracy of Heart Plaque diseasediagnosis using a Naive Bayes- algorithm approach rather than Least Squares Support Vector Machine. [13] This lays the groundwork to enhance cardiovascular diagnosis. Naive Baye's better accuracy in detecting plaque deposits serves us as a model for future predictive analysis and intervention. That will also set the stage for further advances in ML-focused healthcare diagnostics.

II. Proposed Methodology

Simulation results shows that the proposed method obtain a higher configuration accuracy The statistic tool used was G power making use of a probability of 80% and were analyze the 14 samples. Hardware Requirement: 1920x1080 pixel, generation-2 processor, Ryzen 5 series, 8GB RAM and 512 GB SSD worked as the hardware requirement for the training of datasets and MATLAB program with appropriate libraries & tools as the software requirement. The output has been obtained using the software called MATLAB. [14].

The dataset was downloaded from Kaggle and then the data import to Google Colab Group 1 Next, went through multiple iterations, precision was calculated. We have a total of 20 samples on each group to determine an accuracy score. Group 1: Data preparation from sample preparation here it relates to classification and regression, Naive bayes, supervised learning techniques. It's a non-parametric test. It showed a strong performance, for both: classification and regression [15]. Novel Intensity Feature Based Naive Bayes The Novel Intensity Feature Based Naive Bayes is a two-stage classification process which consists of a learning phase and a prediction step [16]. In the learning stage, we use the training data to train the model, trained to predict the output, and when the model is tested with the test data, it is called the prediction stage.

With regards to group 2, the sample preparation involved downloading a dataset from Kaggle and copying said dataset into Google Colab. Then the precision was calculated for different runs. Across 20 samples per group for the accuracy score In group 2, LSSVM (Least Squares Support Vector Machine, supervised learning algorithm, learning from the data and exploring patterns, showed below, was used for analysis. This kind of approach resembles ones encountered in statistics and statistical modeling. Instead of formulating and solving a convex quadratic programming (QP) problem in the classical sense, we solve a set of linear equations to obtain the solution. In this context, the least-squares SVM classifiers (LSSVM) were presented as a kernel-based learning techniques. LS-SVMs is a member of kernel-based learning methods.

The Heart Plaque disease data collection was downloaded from kaggle. To prepare the dataset for training and test, and to perform preprocessing on the dataset. You perform data cleaning whenever you process data like Remove missing and null values, Impute mean or median values, and Scale data. Methods Through: KNN → And LSSVM → Images are input to the preprocessed data 70% of the data is trained out of the whole sample size and 30% data is tested so as to give the most correct result and detection.

This Heart Plaque doctor/g.. Pneumonia pneumonia is a consolidated data set in which each operating system's corresponding overlapping data segment is merged and the name of each operating system is assigned to each operating system's corresponding label file. This study contains 427 healthy person samples and 232 patients with cardiac plaque disease samples. I would be subscribed you people who found an interesting database to work with (12 fields and 659 records) They cover pregnant women, family histories, blood pressures, cholesterol levels, ages, smoking and both ekg and blood sugar levels and outcomes of 659 people.

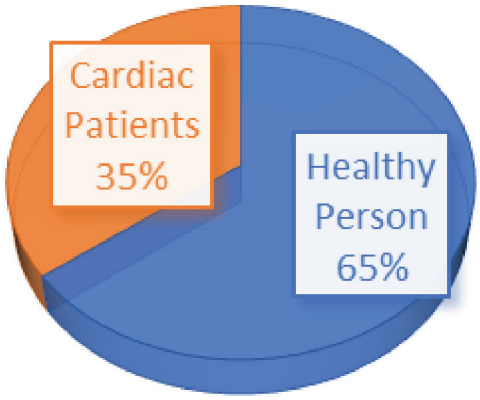

Fig. 3. Patient records

III. Statistical Analysis

Data were analyzed using SPSS software version 21. The proposed and current algorithms were executed 10 times and a maximum single cycle of 20 samples done and the accuracy obtained. Significant values were determined through an independent sample t-test. The independent variables will be Pregnancy, family history, blood pressure, cholesterol, age, smoking, ECG, and blood sugar, while the dependent variable will be the outcome. These data were used to generate a predictive model of the likelihood of suffering from cardiac disease. [20]

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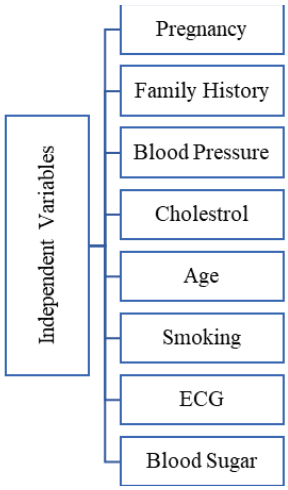

Fig. 4. Independent variables in data analysis

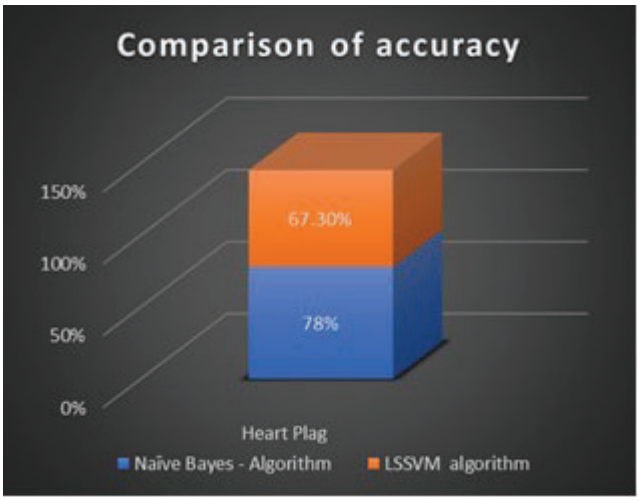

Fig. 5. Comparison of accuracy

TABLE I. COMPARISON OF ACCURACY BETWEEN NAIVE BAYES ALGORITHM AND LSSVM ALGORITHM.

Dataset	Naive Bayes-algorithm		LSSVM algorithm.	
Heart Plaque	Accuracy	78 %	Accuracy	67.3 %

We can deduce that ALL of the LSSVM and Naive Bayes algorithm can be used for the determination of cardiac plaque as seen in figure 4, but the Naive Bayes method shows improved results compared to LSSVM algo (78% vs 67.3% accuracy). In the graphical representation below, it is clearly shown how Naive Bayes starts performing better than LSSVM in producing valid cardiac plaques point predictions.

TABLE II. INDEPENDENT SAMPLE T-TEST

		Levene's test for Equality of Variances		T-test for Equality of Means						
		F	sig.	T	df	Sig. (2-tailed	Mean Diff	Std. Error Diff	Lower	Upper
Accuracy	Equal variances assumed	3.027	0.02	1.517	38	0.000	0.838	0.552	-0.02	0.195
	Not equal variances assumed			1.517	36.02	0.000	0.838	0.552	-0.02	0.195

IV. RESULTS

The Naive Bayes algorithm achieved an accuracy of 78%, and the LSSVM algorithm accuracy 67.3%. The analysis was conducted on a training dataset (n=489) and a test dataset (n=277) using a dataset of 659 samples. A range of tools were used for data analysis. Firstly, version 21 of IBM SPSS was used to analyze data and also accuracy scores and G power statistical tool and independent samples t-tests. Overall, it is fascinating to note in detail, that this topic highlights the impact of using Naive bayes on effectively predicting high-risk patients, by prediction analysis and monitoring of heart plaque formations (diagnostic data of patients).

Experimental resultsBased on the above discussion of the Naive Bayes-algorithm and the LSSVM, we present the experimental results in table 1 for the Heart Plaque disease data set. The Naive Bayes-algorithm was at an accuracy of 78%, and the LSSVM method got an accuracy of 67%. In Table2 individuals can see that the Naive Bayes-algorithm had an average accuracy of 78%, while the LSSVM results were only 67%. We got the 0.194 and 0.152 std deviation of Novel intensity feature Naive Bayes-algorithm and LSSVM algorithm respectively. For the Naive bayes the std error mean was 0.043, while for the LSSVM it was 0.034. The information is seen into the laws: Generally, the average accuracy of Naive Bayes-algorithm became better than comparable quality LSSVM algorithm.

With a 5% significance level the hypothesis is accepted since significance value < 0.05.

V. DISCUSSION

The Novel ensemble algorithm was a better classification model than the LSSVM approach (p=0.01, Independent sample Test) in identifying patients with heart plaque disease [68]. With a higher accuracy (mean accuracy= 78) compared to LSSVM technique (mean accuracy= 67), Naive Bayes algorithm clearly stands up to be the winner. The datatypes cover a range of attributes in an attempt to mirror disease state as well as normal/abnormal states in humans. Since its performance is better than that of the LSSVM, the Naive Bayes Transform is used.

[17] showed that boosting improves the Naive Bayes [8] method, while boost also enhanced the LSSVM. So some improvements can be captured by these two methods and accuracy of other other algorithms also. However, with and

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without bagging, both LSSVM and Naive Bayes algorithm yielded a total accuracy of 67 % only. It happened when the new bagging strategies were implemented. to evaluated classification algorithms (used some classifiers such accuracy, sensitivity, recall and specificity as performance indicator (the researcher [18]). Out of these, based on data up to October 2023 [19-20], the researchers presented a Novel intensity feature-based Naive Bayes algorithm which improves by 75% upon boosting. According to the results, they approached conclusion with two methods the Naive Bayes algorithm technique and LSSVM and it is quite clear that the Naive Bayes algorithm technique is more effective and has proven to perform better than which by a margin of 69 percent. The author evaluated the classifying algorithms, used dense NET, and different machines learning algorithms: their conclusion is that multiple machine learning algorithms have a performance advantage of 87 % compared to the algorithms employed[52]. It would be extremely tightly correlated and can't be folded back into our business logic. All those weights make it difficult to try to understand and analyse the model do and so many individual effects that we cannot do (in a few calibrations to the model). This is something that can be improved in the future by collecting more data.

Hugging Face's problem is also about training time — if you use large datasets, you will need a long time or a lot of expensive GPUs. The Naive Bayes model could also potentially have better accuracy using a multiclass Naive Bayes method on a dataset that could provide a more robust training set.

VI. Conclusion

Technology is a bridge that saves lives, allowing the work of early detection of heart attacks and strokes, the top two killers, to continue. This reiterates the necessity of using ML algorithms in going towards precise diagnosis of such diseases, our work too encourages preventive health care. The Naive Bayes surprisingly secures a notable accuracy success point with 78% proper coverage whereas the LSSVM technique with only 67.3%. A comparison with SVM technique illustrated that Naïve Bayes technique was most favorable for the building on accurate prediction model. Notably this breakthrough represents a paradigm shift by surpassing previous accuracy benchmarks set by other researchers on the same datasets. Beyond just improving upon prior results, our approach also raises the bar for predictive analytics on heart health by obviating the traditional degradation of knowledge preparation and check units. Instead, with this new research we can dream of the day when automation and better coded algorithms for early detection SAVE LIVES AND CAN SHAPE CARDIOVASCULAR CARE. If the Naive Bayes method was a mere statistical victory, it’s in effect the promise of what’s to follow — a world where every pulse will be recorded with unprecedented precision.

References

[1] Chen, JT. et al. (2022). Detection and Classification of Coronary Artery Plaques in Coronary Computed Tomography Angiography Using 3D CNN. In: Puyol Antón, E., et al. Statistical Atlases and Computational Models of the Heart. Multi-Disease, Multi-View, and Multi-Center Right Ventricular Segmentation in Cardiac MRI Challenge. STACOM 2021. Lecture Notes in Computer Science(), vol 13131. Springer, Cham. https://doi.org/10.1007/978-3-030-93722-5_23

[2] Balocco, S., González, M., Ñanculef, R., Radeva, P., Thomas, G. (2018). Calcified Plaque Detection in IVUS Sequences: Preliminary Results Using Convolutional Nets. In: Hernández Heredia, Y., Milián Núñez, V., Ruiz Shulcloper, J. (eds) Progress in Artificial Intelligence and Pattern Recognition. IWAIPR 2018. Lecture Notes in Computer Science(), vol 11047. Springer, Cham. https://doi.org/10.1007/978-3-030-01132-1_4

[3] Hari Priya, A., Vanithamani, R. (2019). Atherosclerotic Plaque Detection Using Intravascular Ultrasound (IVUS) Images. In: Pandian, D., Fernando, X., Baig, Z., Shi, F. (eds) Proceedings of the International Conference on ISMAC in Computational Vision and Bio-Engineering 2018 (ISMAC-CVB). ISMAC 2018. Lecture Notes in Computational Vision and Biomechanics, vol 30. Springer, Cham. https://doi.org/10.1007/978-3-030-00665-5_132

[4] Aggarwal, A., Mittal, R., Gupta, S., Mittal, A. "Internet of things driven perceived value co-creation in smart cities of the future: a PLS-SEM based predictive model." Journal of Computational and Theoretical Nanoscience 16.9 (2019): 4053-4058.

[5] Singh, J.P., Chand, P.K., Mittal, A., Aggarwal, A. "High-performance work system and organizational citizenship behaviour at the shop floor." Benchmarking: An International Journal 27.4 (2020): 1369-1398.

[6] Shalender, Kumar, and Rajesh Kumar Yadav. "Strategic flexibility, manager personality, and firm performance: The case of Indian Automobile Industry." Global Journal of Flexible Systems Management 20 (2019): 77-90.

[7] Goyal, J., Singh, M., Singh, R., Aggarwal, A. "Efficiency and technology gaps in Indian banking sector: Application of meta-frontier directional distance function DEA approach." The Journal of finance and data science 5.3 (2019): 156-172.

[8] Manohar, S., Mittal, A., Marwah, S. "Service innovation, corporate reputation and word-of-mouth in the banking sector: A test on multigroup-moderated mediation effect." Benchmarking: An International Journal 27.1 (2020): 406-429.

[9] Mittal, A., Dhiman, R., Lamba, P. "Skill mapping for blue-collar employees and organisational performance: A qualitative assessment." Benchmarking: An International Journal 26.4 (2019): 1255-1274.

[10] Luo, G. et al. (2019). A Deep Reinforcement Learning Framework for Frame-by-Frame Plaque Tracking on Intravascular Optical Coherence Tomography Image. In: Shen, D., et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2019. MICCAI 2019. Lecture Notes in Computer Science(), vol 11764. Springer, Cham. https://doi.org/10.1007/978-3-030-32239-7_2

[11] Mahadevi, C., Sivakumar, S. (2020). Performance of Coronary Plaque Feature Extraction and Identification of Plaque Severity for Intravascular Ultrasound B-Mode Images. In: B. R., P., Thenkanidiyoor, V., Prasath, R., Vanga, O. (eds) Mining Intelligence and Knowledge Exploration. MIKE 2019. Lecture Notes in Computer Science(), vol 11987. Springer, Cham. https://doi.org/10.1007/978-3-030-66187-8_21

[12] Zhang, C., He, J., Wang, W., Yang, S., Zhang, Y. (2019). A Real-Time Method for Marking the Extent of a Lipid Plaque Based on IV-OCT Imaging. In: Sun, F., Liu, H., Hu, D. (eds) Cognitive Systems and Signal Processing. ICCSIP 2018. Communications in Computer and Information Science, vol 1005. Springer, Singapore. https://doi.org/10.1007/978-981-13-7983-3_9

[13] Aher, C.N., Jena, A.K. (2023). Heart Disease Detection from Gene Expression Data Using Optimization Driven Deep Q-Network. In: Bhateja, V., Yang, XS., Chun-Wei Lin, J., Das, R. (eds) Intelligent Data Engineering and Analytics. FICTA 2022. Smart Innovation, Systems and Technologies, vol 327. Springer, Singapore. https://doi.org/10.1007/978-981-19-7524-0_53

[14] Sai Kiran, K.V., Azman, M., Nandu, E., Sai Prakash, S.K.L.V. (2021). Real-Time Detection and Prediction of Heart Diseases from ECG Data Using Neural Networks. In: Gunjan, V.K., Zurada, J.M. (eds) Proceedings of International Conference on Recent Trends in Machine Learning, IoT, Smart Cities and Applications. Advances in Intelligent Systems and Computing, vol 1245. Springer, Singapore. https://doi.org/10.1007/978-981-15-7234-0_9

[15] Yadav, R., Dahiya, V., Vashisth, S. (2023). The Automation of Different Plaque Shapes to Compute Influence of Blood Flow and Wall Shear Stress in Stenosed Artery Using ANSYS. In: Khosla, A., Kolhe, M. (eds) Renewable Energy Optimization, Planning and Control. Studies in Infrastructure and Control. Springer, Singapore. https://doi.org/10.1007/978-981-19-8963-6_33

[16] Kumar, R., Aggarwal, Y., Nigam, V.K. (2021). Biomarkers and Heart Rate Variability in the Prognosis of Cardiovascular Disease: A Perspective. In: Mukherjee, M., Mandal, J., Bhattacharyya, S., Huck, C.,

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Biswas, S. (eds) Advances in Medical Physics and Healthcare Engineering. Lecture Notes in Bioengineering. Springer, Singapore. https://doi.org/10.1007/978-981-33-6915-3_7

[17] Liapi, G.D., Kyriacou, E., Loizou, C.P., Panayides, A.S., Pattichis, C.S., Nicolaides, A.N. (2022). Deep Learning-Based Segmentation of the Atherosclerotic Carotid Plaque in Ultrasonic Images. In: Maglogiannis, I., Iliadis, L., Macintyre, J., Cortez, P. (eds) Artificial Intelligence Applications and Innovations. AIAI 2022 IFIP WG 12.5 International Workshops. AIAI 2022. IFIP Advances in Information and Communication Technology, vol 652. Springer, Cham. https://doi.org/10.1007/978-3-031-08341-9_16

[18] Denzinger, F. et al. (2021). Coronary Plaque Analysis for CT Angiography Clinical Research. In: Palm, C., Deserno, T.M., Handels, H., Maier, A., Maier-Hein, K., Tolxdorff, T. (eds) Bildverarbeitung für die Medizin 2021. Informatik aktuell. Springer Vieweg, Wiesbaden. https://doi.org/10.1007/978-3-658-33198-6_53

[19] Bazilevych, K., Butkevych, M., Dotsenko, N., Padalko, H. (2023). Bagging Model for Heart Stroke Detection. In: Nechyporuk, M., Pavlikov, V., Kritskiy, D. (eds) Integrated Computer Technologies in Mechanical Engineering - 2022. ICTM 2022. Lecture Notes in Networks and Systems, vol 657. Springer, Cham. https://doi.org/10.1007/978-3-031-36201-9_24

[20] Cao, X., Zheng, J., Liu, Z., Jiang, P., Gao, D., Ma, R. (2021). Improved U-Net for Plaque Segmentation of Intracoronary Optical Coherence Tomography Images. In: Farkaš, I., Masulli, P., Otte, S., Wermter, S. (eds) Artificial Neural Networks and Machine Learning – ICANN 2021. ICANN 2021. Lecture Notes in Computer Science, vol 12893. Springer, Cham. https://doi.org/10.1007/978-3-030-86365-4_48

[21] Denzinger, F. et al. (2020). Deep Learning Algorithms for Coronary Artery Plaque Characterisation from CCTA Scans. In: Tolxdorff, T., Deserno, T., Handels, H., Maier, A., Maier-Hein, K., Palm, C. (eds) Bildverarbeitung für die Medizin 2020. Informatik aktuell. Springer Vieweg, Wiesbaden. https://doi.org/10.1007/978-3-658-29267-6_42

[22] Liu, J., Jin, C., Feng, J., Du, Y., Lu, J., Zhou, J. (2019). A Vessel-Focused 3D Convolutional Network for Automatic Segmentation and Classification of Coronary Artery Plaques in Cardiac CTA. In: Pop, M., et al. Statistical Atlases and Computational Models of the Heart. Atrial Segmentation and LV Quantification Challenges. STACOM 2018. Lecture Notes in Computer Science(), vol 11395. Springer, Cham. https://doi.org/10.1007/978-3-030-12029-0_15

[23] Liu, M., Sun, Y., Chen, Y. (2020). Detection of Atherosclerotic Lesions Based on Molecular Communication. In: Chen, Y., Nakano, T., Lin, L., Mahfuz, M., Guo, W. (eds) Bio-inspired Information and Communication Technologies. BICT 2020. Lecture Notes of the Institute for Computer Sciences, Social Informatics and Telecommunications Engineering, vol 329. Springer, Cham. https://doi.org/10.1007/978-3-030-57115-3_19

[24] Onhirun, N., Charoenporn, P. (2022). Automatical IVUS Image Segmentation and Detection Using BLOB Analysis. In: Meesad, P., Sodsee, S., Jitsakul, W., Tangwannawit, S. (eds) Proceedings of the 18th International Conference on Computing and Information Technology (IC2IT 2022). IC2IT 2022. Lecture Notes in Networks and Systems, vol 453. Springer, Cham. https://doi.org/10.1007/978-3-030-99948-3_1

[25] Archana, K.V., Vanithamani, R. (2019). Diagnosis of Coronary Artery Diseases and Carotid Atherosclerosis Using Intravascular Ultrasound Images. In: Hemanth, J., Silva, T., Karunananda, A. (eds) Artificial Intelligence. SLAAI-ICAI 2018. Communications in Computer and Information Science, vol 890. Springer, Singapore. https://doi.org/10.1007/978-981-13-9129-3_20

[26] Ma, X., Luo, G., Wang, W., Wang, K. (2021). Transformer Network for Significant Stenosis Detection in CCTA of Coronary Arteries. In: de Bruijne, M., et al. Medical Image Computing and Computer Assisted Intervention – MICCAI 2021. MICCAI 2021. Lecture Notes in Computer Science, vol 12906. Springer, Cham. https://doi.org/10.1007/978-3-030-87231-1_50

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